

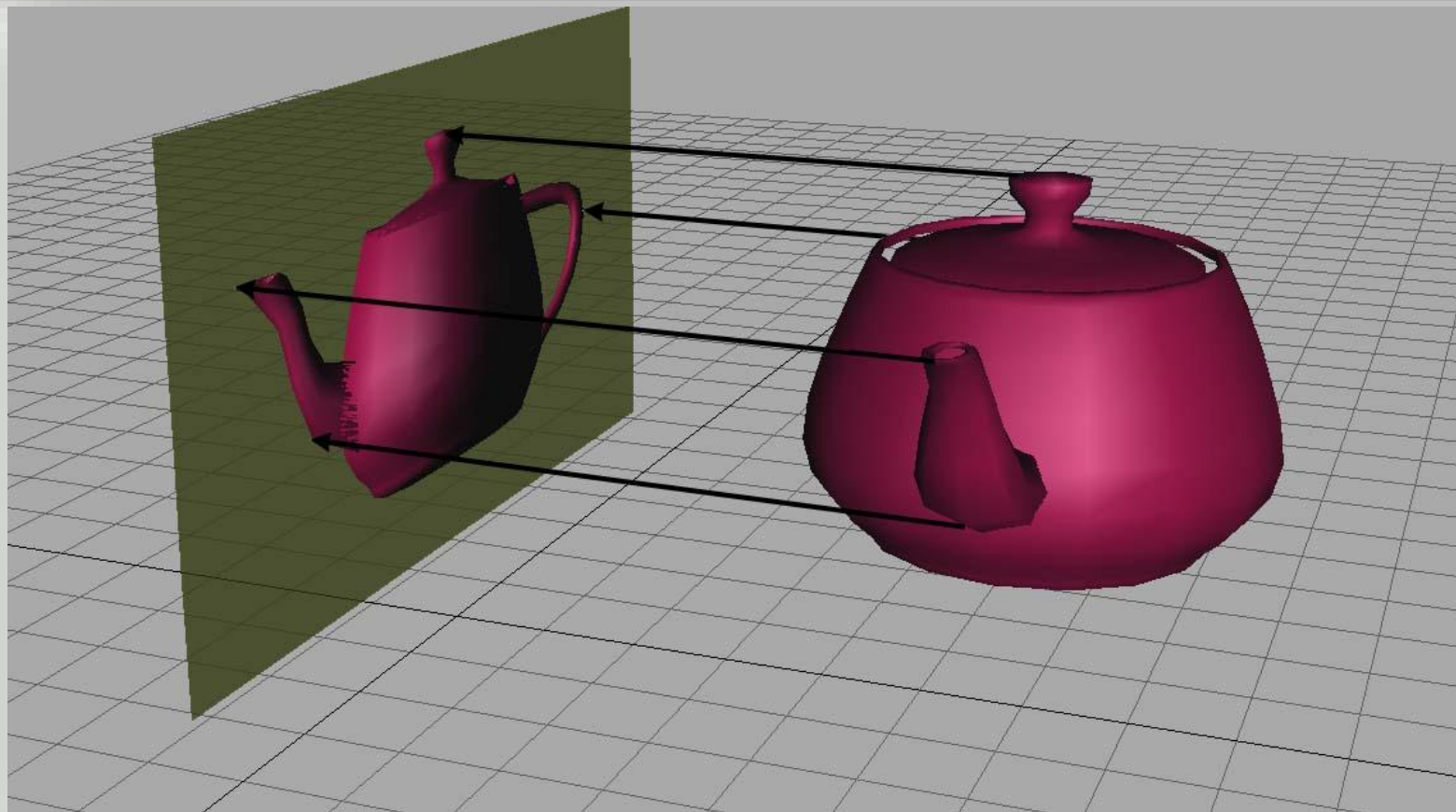
Section 6.4:

4×4 Matrices and Perspective Projection

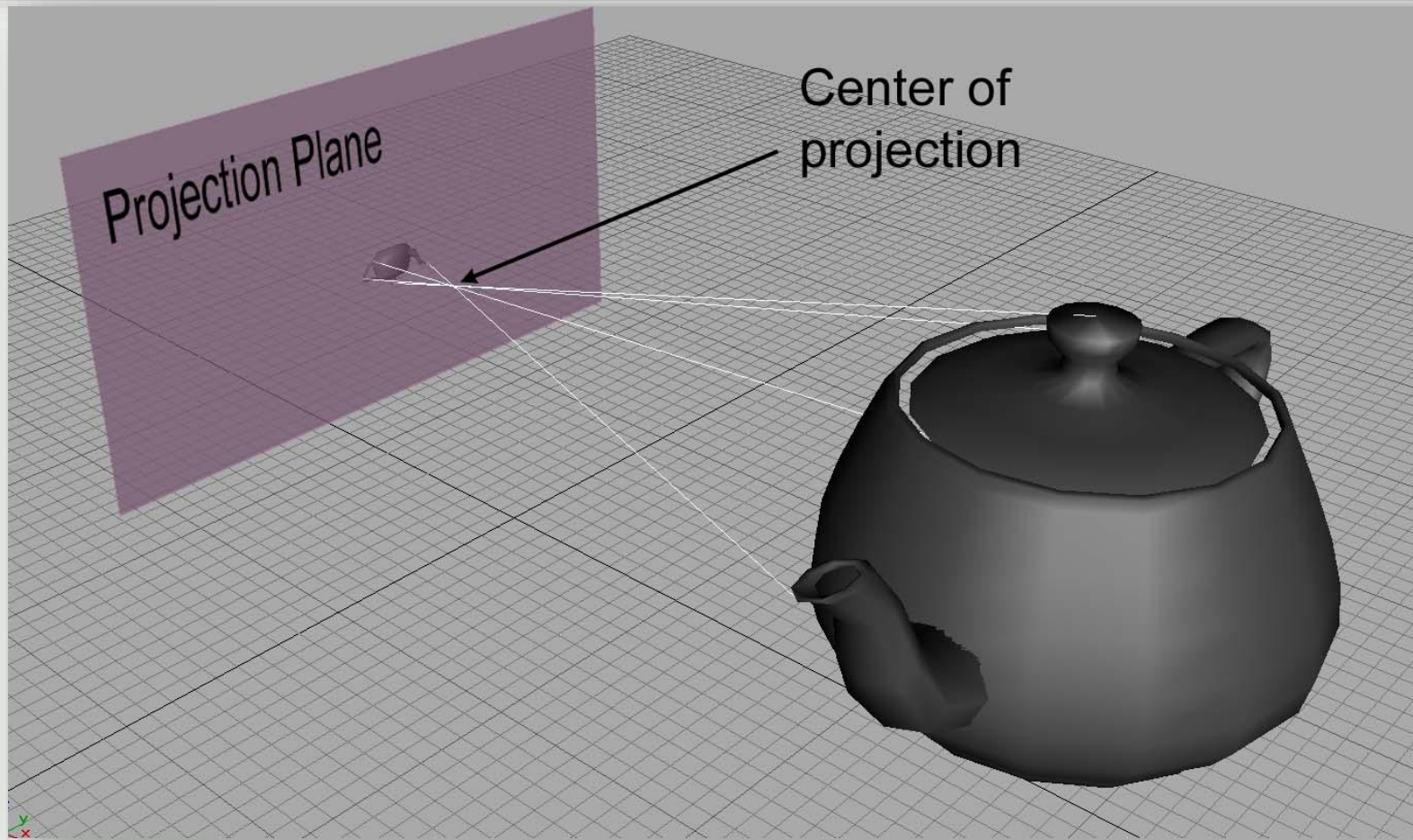
Orthographic Projection

- Orthographic projection has *parallel projectors*.
- The projected image is the same size no matter how far the object is from the projection plane.
- We want objects to get smaller with distance.
- This is known as *perspective foreshortening*.

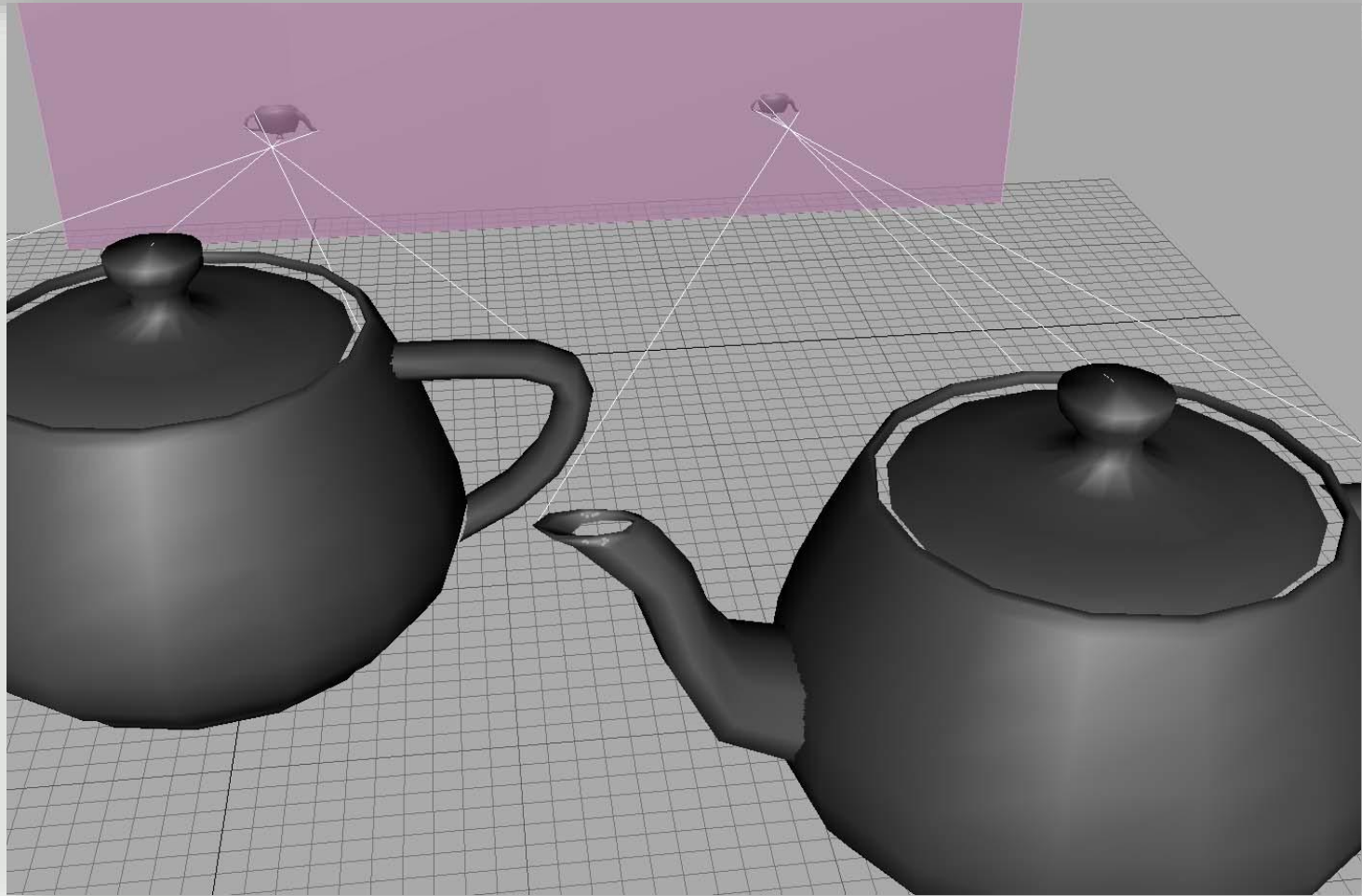
Orthographic Projection



Perspective Projection

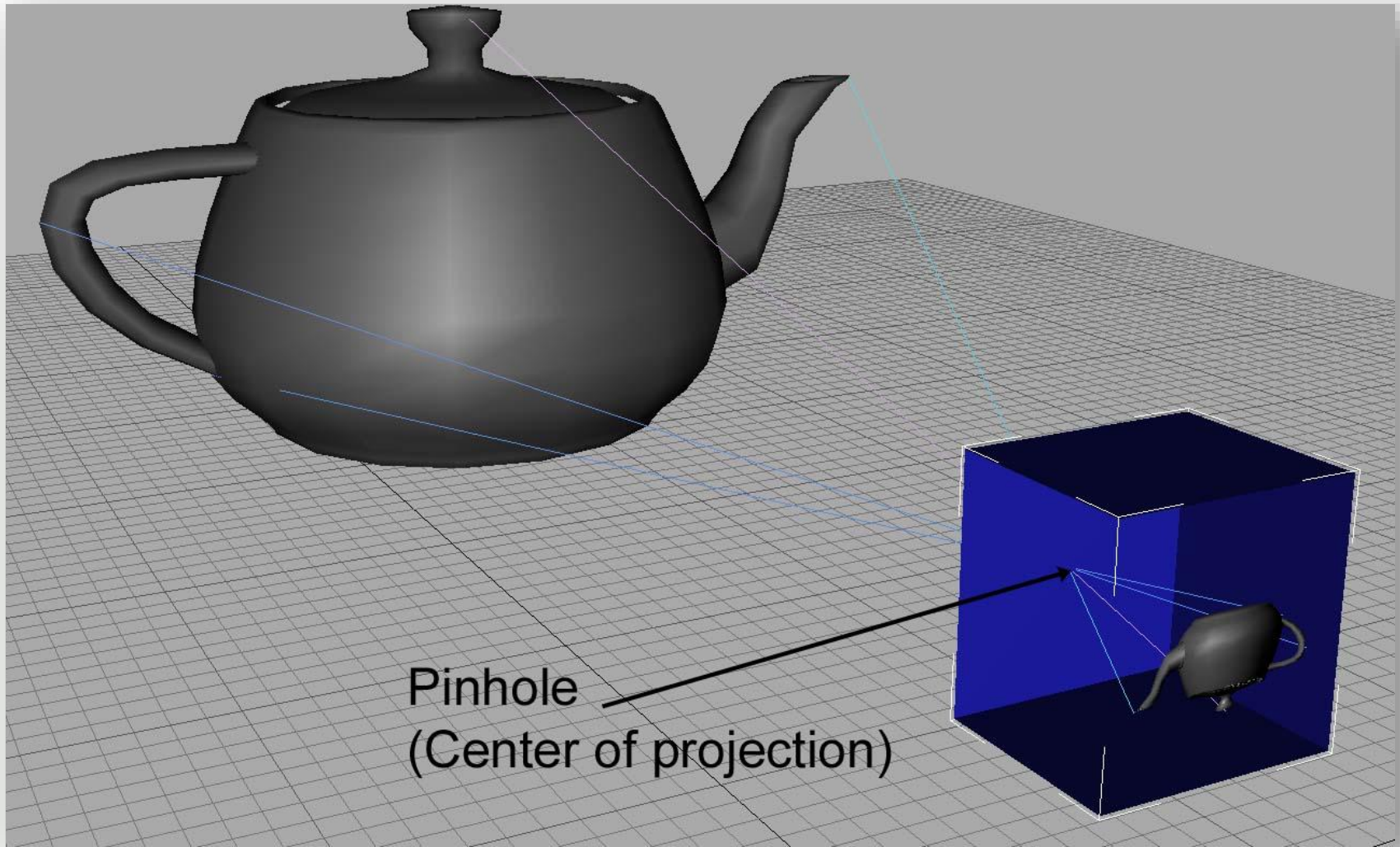


Perspective Foreshortening



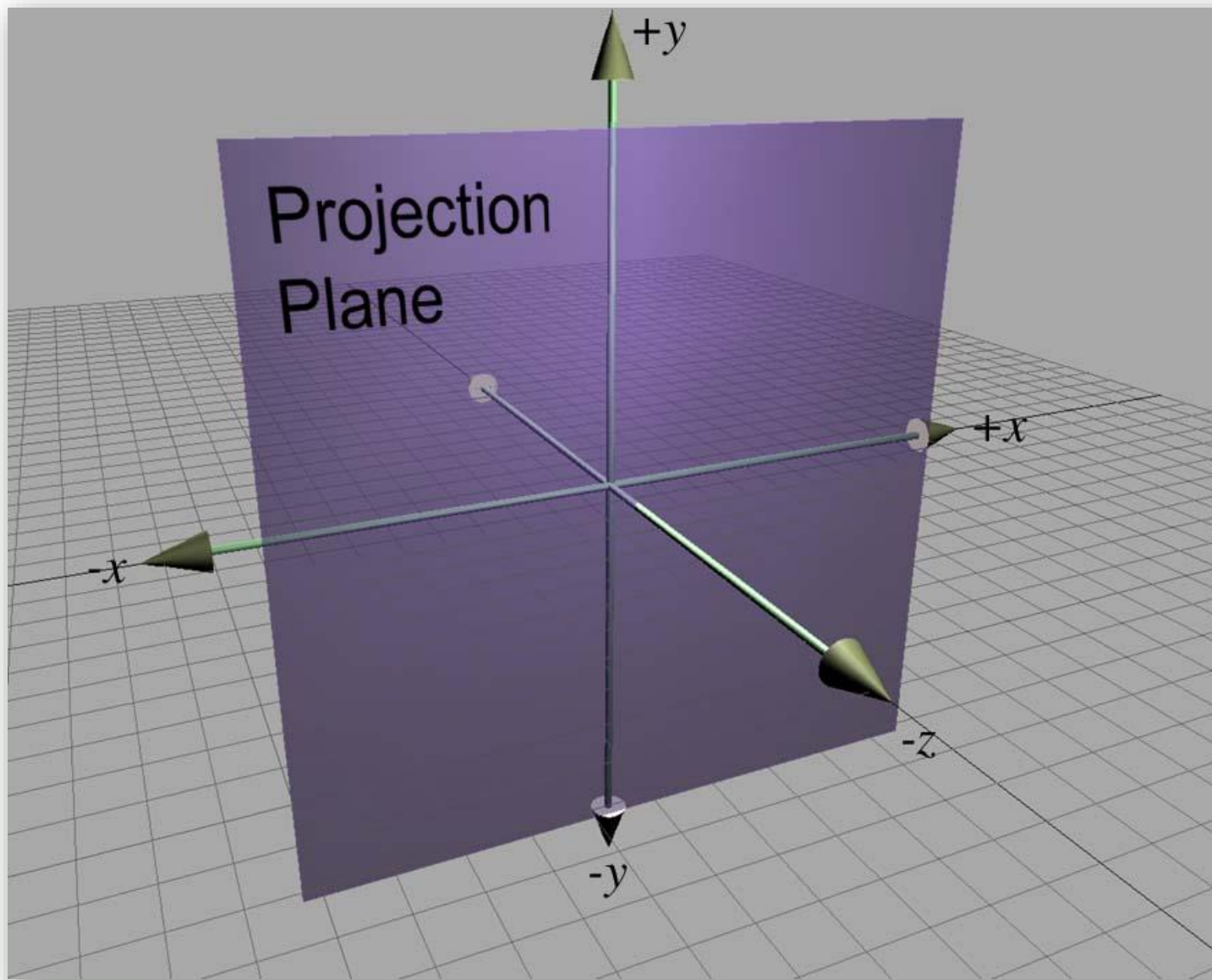
The Pinhole Camera

- The math is based on a *pinhole camera*.
- Take a closed box that's very dark inside.
- Make a pinhole.
- If you point the pinhole at something bright, an image of the object will be projected onto the back of the box.
- That's kind of how the human eye works too.



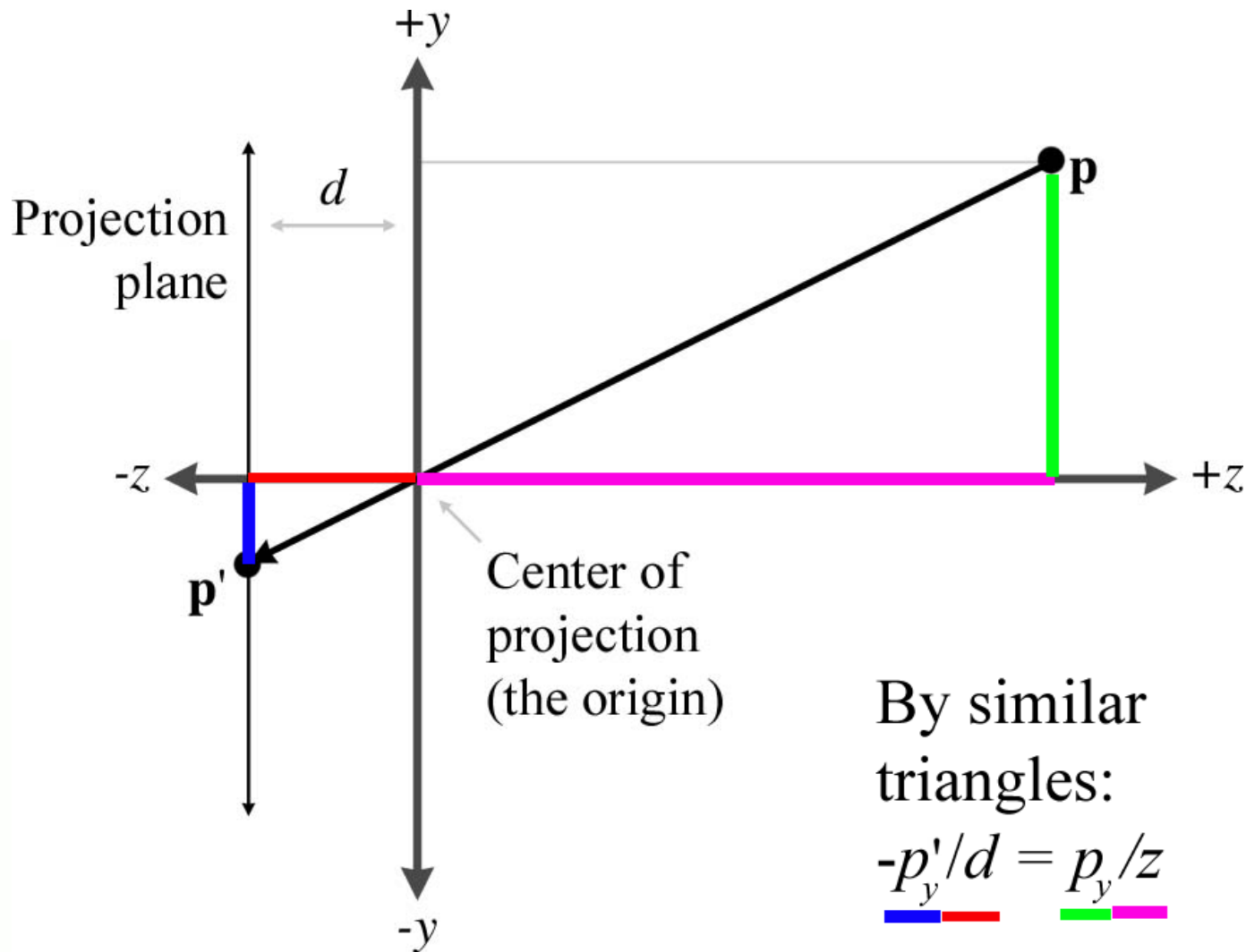
Projection Geometry

- Let's project on a plane parallel to the x - y plane.
- Choose a distance d from the pinhole to the projection plane, called the *focal distance*.
- The pinhole is called the *focal point*.
- Put the focal point at the origin and the projection plane at $z = -d$.
- (Remember the concept of *camera space*?)



Do the Math

- View it from the side.
- Consider where a point $\mathbf{p}$ gets projected onto the plane – at a point $\mathbf{p}'$
- Start with the y coordinate for now.



Finding p'

From the previous slide, by similar triangles:

$$\frac{-p'_y}{d} = \frac{p_y}{z} \implies p'_y = \frac{-dp_y}{z}$$

Same for the x-axis:

$$p'_x = \frac{-dp_x}{z}$$

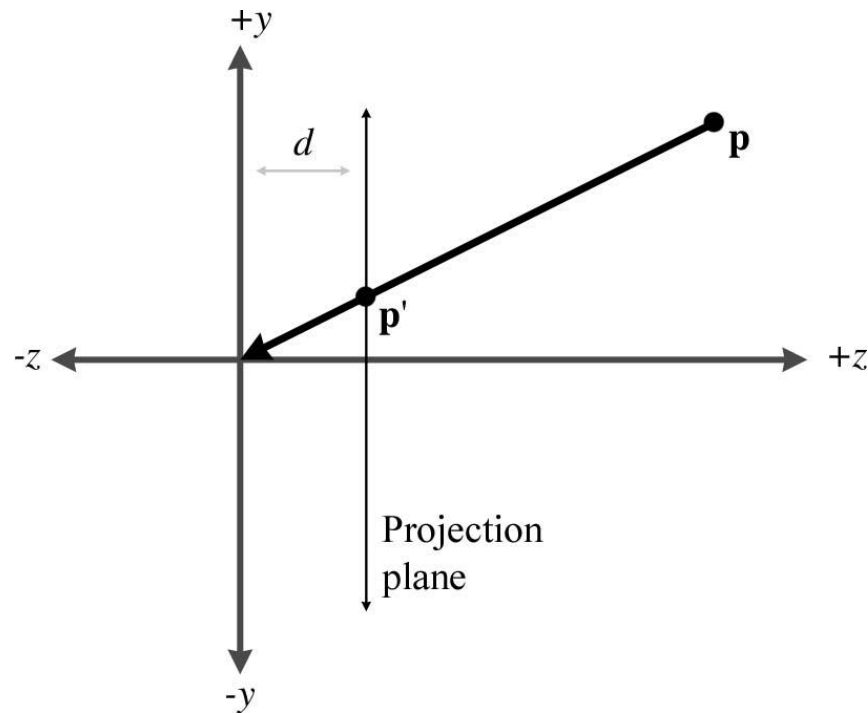
Result

- All projected points have a z value of $-d$.
- Therefore $\mathbf{p}$ is projected onto $\mathbf{p}'$ like this:

$$\mathbf{p} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \implies \mathbf{p}' = \begin{bmatrix} x' \\ y' \\ z' \end{bmatrix} = \begin{bmatrix} -dx/z \\ -dy/z \\ -d \end{bmatrix}$$

In Practice

In practice we move the projection plane in front of the focal point.



Accentuate the Positive

Doing so removes the annoying minus signs.
This:

$$\mathbf{p} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \implies \mathbf{p}' = \begin{bmatrix} x' \\ y' \\ z' \end{bmatrix} = \begin{bmatrix} -dx/z \\ -dy/z \\ -d \end{bmatrix}$$

Becomes this:

$$\mathbf{p}' = \begin{bmatrix} x' \\ y' \\ z' \end{bmatrix} = \begin{bmatrix} dx/z \\ dy/z \\ d \end{bmatrix}$$

Projection Using 4D Matrix

- We can actually do this with a 4D homogenous matrix.
- First manipulate $\mathbf{p}'$ to have a common denominator:

$$\begin{aligned}\mathbf{p}' &= [dx/z \quad dy/z \quad d] \\ &= [dx/z \quad dy/z \quad dz/z] \\ &= (d/z) [x \quad y \quad z]\end{aligned}$$

Entering the 4th Dimension

- To multiply by d/z , divide by z/d .
- Instead of dividing by z/d , make that our w coordinate:

$$[x \ y \ z \ z/d]$$

- We need a 4x4 matrix that transforms an “ordinary” point $[x \ y \ z \ 1]$ into this.

The Projection Matrix

$$\begin{bmatrix} x & y & z & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1/d \\ 0 & 0 & 0 & 0 \end{bmatrix} \\ = \begin{bmatrix} x & y & z & z/d \end{bmatrix}$$